

ATHARV ASHOK PATIL

M.Sc. Data Science Student

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[GitHub](#) | [LinkedIn](#) | English (IELTS 7.5 / C1) | German (B1, in progress)

PROFILE SUMMARY

M.Sc. Data Science student at TU Braunschweig with 2.5 years of industry experience in machine learning, real-time signal processing, and AI systems. Currently working at Fraunhofer IWU on synthetic data generation, diffusion models, and computer vision for industrial defect inspection. Seeking an internship and/or Master's thesis opportunity in Computer Vision, Applied AI, or Machine Learning.

EDUCATION

Technische Universität Braunschweig

Master of Science, Data Science

Apr 2023 – Sept 2027 (Expected)

Braunschweig, Germany

- Grade: 2.1
- Relevant Coursework: ML & Deep Learning, Statistical Data Analysis, Numerical Methods, Optimization.

Vidyalankar Institute of Technology, University of Mumbai

Bachelor of Engineering, Information Technology

Jun 2018 – May 2022

Mumbai, India

- CGPA: 8.72/10 (~ German Grade: 1.6)
- Relevant Coursework: Data Analysis, DBMS, Software Engineering, Programming, Python.

WORK EXPERIENCE

Fraunhofer IWU – Prüftechnik und Automation

Generative AI and Computer Vision – HiWi

Jun 2026 – Present

Chemnitz, Germany

- Built an end-to-end synthetic defect generation pipeline (Python, OpenCV, PyTorch, Stable Diffusion) for industrial surface inspection.
- Benchmarked three texture synthesis methods across six metal finishes, reaching an LPIPS score of 0.13.
- Resolved a blending-artifact issue with zone-based compositing and used TPS warping to generate 20+ defect variants per sample.
- Fine-tuned Stable Diffusion with LoRA, updating just 0.5% of parameters, for precise defect edits.
- Built a PatchCore-based evaluation framework to quantify the relationship between synthetic defect fidelity and downstream detectability.

Ingenieurbüro: Dr.-Ing. Eugen Koch

Biomedical Signal Processing & Data Science - Werkstudent

Nov 2023 – Jan 2026

Braunschweig, Germany

Project: Real-Time Smart Ventilator Triggering System for Neonatal (Premature) Babies

- Engineered a high-frequency signal processing pipeline (SciPy, NumPy) with Savitzky-Golay filtering and adaptive normalization, achieving **<15ms latency** for respiratory detection (**92% precision, 81% recall**).
- Developed a hybrid control algorithm with a fail-safe apnea state machine, **reducing ventilator asynchrony by 61.8%**.
- Leveraged **Azure ML HyperDrive** for **hyperparameter tuning** of detection thresholds.
- Built real-time diagnostic dashboard (PyQtGraph) displaying live metrics (Precision, Recall, MAD).

Institut für Philosophie, TU Braunschweig

IT-HiWi & DV-Koordinator

Mar 2024 – Aug 2026

Braunschweig, Germany

- Administered Ubuntu file servers and Samba-based network storage, reducing downtime by 50% through proactive maintenance.
- Supported system hardening and cybersecurity initiatives, improving institutional data protection practices.

PROJECTS

Facial Emotion Recognition: MLP vs CNN & Label Quality

Jan 2026 – Mar 2026

- Compared MLP vs CNN on FER-2013, showing **~17pp gain from spatial inductive bias**.
- Improved performance using **FER+ soft labels: MLP +18.3pp (48.6% to 66.9%), CNN 82.2%** overall.
- Built training pipelines in PyTorch with soft cross-entropy, class balancing, and LR scheduling.

Predictive Modeling for Airline Network Delays

Oct 2025 – Dec 2025

- Built an optimized ETL pipeline for US BTS flight data, reducing memory usage by **50%** and preventing system crashes.
- Applied **stratified sampling and weather-based feature engineering** to predict NAS delay and display insights on a **Streamlit dashboard**.

Real-Time Traffic Object Detection using YOLOv8 (Deep Learning)

Oct 2025 – Nov 2025

- Fine-tuned YOLOv8n on the Pascal VOC dataset for real-time traffic object detection, to identify key traffic entities (*cars, buses, bicycles, motorbikes*) in images and video streams.
- Achieved strong benchmark performance on the test set with **mAP@0.5 = 0.836** and **mAP@0.5:0.95 = 0.631**, demonstrating accurate object localization and classification.
- Built a real-time webcam inference pipeline achieving **20–50 FPS** on consumer-grade hardware while maintaining **high IoU-based detection accuracy**.

Financial Time-Series Forecasting

April 2024 – Dec 2024

- Built a cloud-native Forex trading pipeline on Azure ML using **tsfresh for automated feature extraction** and a Random Forest classifier with PCA for directional market prediction.
- Applied **Walk-Forward Validation and dynamic hedging strategies** (EUR/USD to USD/CHF, EUR/USD to USD/JPY) across **daily and 4-hr timeframes**, analyzing equity growth and drawdowns in backtesting.
- Deployed an **automated MT5 trading bot** with break-even and early stopping logic, enabling risk-aware live execution with adaptive hedging.

SKILL SET

- **Programming** : Python, C++, R, Java, C#, HTML, CSS
- **ML & AI** : NumPy, Pandas, SciPy, Scikit-learn, PyTorch, TensorFlow, NLP (spaCy, NLTK)
- **Computer Vision** : OpenCV, YOLOv8, Image Processing and Segmentation, FFT/Spectral Analysis
- **Generative AI** : Stable Diffusion, Hugging Face Diffusers, LoRA (PEFT)
- **Cloud & MLOps** : Azure ML (Pipelines, HyperDrive), Docker, Linux, Git, SQL, CI/CD
- **Visualization** : Streamlit, Power BI, Tableau, Matplotlib, PyQtGraph

CERTIFICATIONS

- [Introduction to Generative AI](#) - GoogleSkills
- [Introduction to LLMs](#) - GoogleSkills
- [Managing Big Data with MySQL](#) - Duke University (Coursera)
- [Python for Data Analysis, Data Science & ML](#) - Udemy
- [Power BI - Data Analytics Essentials](#) - Udemy